

Computer Systems

COMP1071: Durham University

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Part I

Digital Electronics

1 Circuits

1.1 Transistors

Definition 1.1.1: Transistors are electronically controlled switches that turn on or off when voltage or current is applied to a control terminal.

Definition 1.1.2: A nMOS transistor is a sandwich of layers of conducting and insulating material. The gate is a conducting layer, on top of an insulating layer, on top of the substrate which is a semi-conductor. When its gate g is at a high voltage, current is allowed to flow from source to drain, and this is prevented when the gate is at a low voltage.

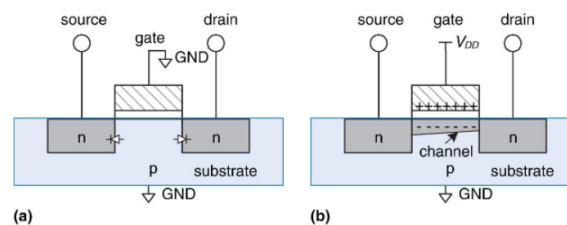


Figure 1: An nMOS transistor

The junction between p-type and n-type silicon is a diode. This means the current can only flow from p-type to n-type as they act as anodes and cathodes respectively.

Definition 1.1.3: A pMOS transistor is the opposite of a nMOS transistor - it allows current to flow when the gate is at low voltages, and prevents current flow at high voltages.

Definition 1.1.4: The static discipline states that if inputs to transistors meet valid thresholds, then the system must guarantee outputs will meet valid output thresholds and will not fall into the forbidden zone (outside of the accepted range for high/low voltages).

Definition 1.1.5: Moore's Law is usually quoted as "transistor density doubles every two years", or "computer processing power doubles every eighteen months".

1.2 Circuit Design

Definition 1.2.3: Abstraction is used when designing circuits to simplify problems by hiding unimportant details, highlighting the actual logic problem to be solved.

Definition 1.2.4: Another design principle used is discipline. Here, design choices are deliberately restricted to make circuits easier to model, design, and combine.

Definition 1.2.5: When designing complex systems, engineers use the three-y's: hierarchy, dividing a system into clear modules and submodules; modularity, using well-defined functions and interfaces for modules; and regularity, encouraging uniformity so modules can be swapped or reused.

1.3 Combinatorial Circuits

Definition 1.3.1: A combinatorial circuit is a circuit where the output depends purely on the current input.

Definition 1.3.2: An adder circuit takes the contents of two registers and adds them together.

- **Definition 1.3.3:** A half-adder circuit takes two one-bit binary inputs and adds them, returning a sum bit, and a carry bit.
- **Definition 1.3.4:** A full-adder circuit takes three one-bit binary inputs (one of which is a carry bit) and adds them, returning a sum bit, and a carry bit.

Definition 1.3.5: A decoder circuit takes an n -bit binary number and decodes it into 2^n data lines.

Definition 1.3.6: A multiplexor (mux) circuit uses a binary number to select one of a set of inputs to funnel to its output.

Definition 1.3.7: A tristate takes in two bits, an input bit and an enable bit. If the enable bit is a one, the input bit is outputted, if not then the output is left floating.

1.4 Sequential Circuits

Definition 1.4.1: A sequential circuit is a circuit which changes its output depending on its input history, alongside the current input.

Definition 1.4.2: A circuit which has two stable output states for a given input is known as a bistable circuit.

Definition 1.4.3: An sr-latch takes in two inputs, set S and reset R . If set has a pulse of one, then the output remains at one, even after the pulse ends. If reset has a pulse of one, then the output is reset back to zero. This is a bistable circuit.

Definition 1.4.4: A d-latch has two inputs, a data input D and a clock input CLK . The latch only changes to match the input at D when the CLK signal is high.

Definition 1.4.5: A d-type flip-flop works similarly to the d-latch, but only allows the stored output to change on the rising edge of the clock signal.

2 Introduction to Logic

2.1 Functionally Complete Sets

Definition 2.0.1: A functionally complete set is a set of boolean operators which can be used to express all possible truth tables by combining members of the set into a boolean expression. For example, $\{AND, OR, NOT\}$, $\{NOR\}$, and $\{NAND\}$ are all functionally complete sets.

Part II

Machine Architecture

3 MIPS

3.1 Introduction to MIPS

Definition 3.1.1: A MIPS (Microprocessor without Interlocked Pipeline Stages) is 32-bit RISC processor used in embedded systems and consumer electronics. It uses 32-bit words, has around 110 instructions, and 32 general purpose registers.

Definition 3.1.2: Amdahl's Law states that the performance improvement to be gained from using some faster mode of execution is limited by a fraction of the time the faster mode can be used.

3.2 The MIPS Instruction Set

The instruction set uses five addressing modes, these are:

- **Definition 3.2.1:** When all addresses in the instruction are registers, this is register-only addressing.
- **Definition 3.2.2:** When a value is used directly in an instruction, this is immediate addressing.
- **Definition 3.2.3:** Base addressing is when the address of a memory operand is given by a base address and a signed immediate.
- **Definition 3.2.4:** PC-relative addressing takes the current position in memory and jumps a given amount to a new address.
- **Definition 3.2.5:** Pseudo-direct addressing is used by MIPS in place of direct addressing as it does not support 32-bit addresses. As a result, it calculates it from 26 as the two least significant bits are always zeros as all words are multiples of four, and the four most significant bits can also be left as zero.

The MIPS instruction set can be split into three instruction types, these are:

- **Definition 3.2.6:** R-type instructions contain a six-bit opcode, three five-bit registers (two source and one destination), a five-bit shift value, and a six-bit code to identify the function.
- **Definition 3.2.7:** I-type instructions contain a six-bit opcode, two five-bit registers (one source, one destination), and a sixteen-bit immediate.
- **Definition 3.2.8:** J-type instructions contain a six-bit opcode and a twenty-six bit memory address. The compiler combines the two to form a new thirty-two bit address.

3.3 Memory Maps

MIPS uses 32-bit memory addresses across 2^{32} bits (or 4 GB), ranging from 0x0 to 0xFFFFFFFFFC. This address space is split into the four segments:

- **Definition 3.3.1:** The text segment stores the machine language program and can store 256 MB.
- **Definition 3.3.2:** The global data segment stores global variables for program functions to access. These can be accessed using the pointer `$gp`. This segment can store 64 KB of data.
- **Definition 3.3.3:** The dynamic data segment stores data which is dynamically allocated/deallocated through the execution of the program and can store up to 2 GB of data using a stack and a heap.
- **Definition 3.3.4:** The reserved segment is used by the operating system and cannot be directly accessed by the program.

4 The ATmega328P Microcontroller

4.1 Main Technical Characteristics

Definition 4.1.1: A microcontroller is an integrated device consisting of a microprocessor, memory, and an input/output. They are used in embedded systems and store their program in flash memory.

The microcontroller has 32 general purpose 8-bit registers that are copied onto SRAM memory. Registers 26-31 are used in addressing, and as such should be treated as three 16-bit registers, rather than six 8-bit registers.

Definition 4.1.2: The status register is an 8-bit register separate from the other 32. Its eight bits are used as flags.

The microcontroller's pins are separated into 3 groups, B (digital), C (analogue), and D (digital).

Definition 4.1.3: The direction of each pin of a port is controlled by a data direction registers. We say a bit is set when it is 1 and the corresponding pin is output, and that a bit is cleared when it is 0 and the pin is input.

5 Programming in AVR Assembly

5.1 Directives

Definition 5.1.1: The assembly code contains statements called directives. These do not correspond to assembly instructions, but the assembler will take them into account when compiling. In AVR assembly, a directive is preceded by a dot. Some examples include:

- *.equ* assigns a value to a label which can be used in later expressions. This is a constant. (*.equ pi, 3*).
- *.global start* and *.global btnLED* declare the functions *start* and *btnLED* as global.

Part III

Databases

6 Database Design

6.1 The ANSI-SPARC Architecture

Definition 6.1.1: ANSI (American National Standards Institute) and SPARC (The Standards, Planning, and Requirements Committee).

Part IV

Operating Systems

Part V

References

7 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Digital Electronics* and *Machine Architecture* are taught by Ioannis Ivrisimtzis.
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8 Cited Works